

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA**1st Semester B.Tech. (EC/ME/CL) Examination (IT/CE/EE)****University Examination, December 2010.****CL101 Fundamentals of Civil Engineering****Date: 14-12-10, Tuesday****Time: 10:30a.m to 01:30p.m****Maximum Marks:70****Instructions:**

1. Attempt all the questions.
2. Figures to the right indicate full marks.
3. Make suitable assumptions.

SECTION-I

Q - 1 (A) Describe with a neat sketch, the optical square and explain its principle. How it is used in the field. **(04)**

(B) Discuss the importance of planning, scheduling and construction management. **(03)**

Q - 2 (A) Explain the term primary divisions of surveying. List out various modern tools used for surveying and mapping. Explain any one in detail. **(05)**

(B) Describe the temporary adjustment of a prismatic compass. **(03)**

(C) Write short notes on : (Attempt any two) **(06)**

1. Plan & Map
2. Errors in chaining
3. Classification of surveying based on nature of field.

OR

Q - 2 (A) A closed traverse is drawn by five stations A,B,C,D and E taken in anticlockwise order, in order to form a regular polygon. If fore bearing of line AB is 30° . Find fore bearing of other sides. **(05)**

(B) A surveyor measured the distance between two points on the plan drawn to a scale of $1\text{cm} = 40\text{m}$ and result was 468m. Later he discovered that he used scale of $1\text{cm} = 20\text{m}$. Find the true distance between the points. **(03)**

(C) Attempt any two: **(06)**

1. Fundamental principles of surveying.
2. Meridians and dip of needle.
3. Convert W.C.B. to R.B. and R.B. to W.C.B.

i) $N 35^\circ W$	ii) $S 89^\circ W$	iii) 65°
iv) $345^\circ 30'$	v) $112^\circ 30'$	vi) $N 10^\circ E$

Q - 3 (A) The following bearings were observed on a closed compass traverse. Calculate the interior angles and correct them for observed errors. **(05)**

Line	Fore bearings	Back bearings
AB	$190^\circ 45'$	$10^\circ 45'$
BC	$120^\circ 15'$	$300^\circ 15'$
CD	$348^\circ 35'$	$168^\circ 30'$
DE	$339^\circ 05'$	$160^\circ 10'$
EA	$295^\circ 30'$	$114^\circ 30'$

(P.T.O)

(B) Write short notes:

1. Different chains used in surveying.
2. Branches of Civil Engineering.

(05)

(C) Length of survey line measured with a chain was found to be 400m. Calculate true length of a line, If

(04)

1. The length was measured with 30m chain and chain was 15cm too long.
2. The length of chain was 30m in beginning and 30.10m at the end of work.

OR

Q - 3 (A) The following staff readings were taken with dumpy level at every 30m interval. The observed readings were 2.285, 2.265, 1.650, 2.105, 1.960, 1.625, 1.925, 2.050, 1.665, 1.325, 1.690, 2.100, 2.865, 1.625. The instrument was shifted after 2nd, 5th, 7th, 10th and 12th readings. Enter the above readings in a page of level book and calculate reduced level of points, if the first reading was taken with staff held on bench mark of 232.460. Also apply necessary arithmetic check.

(04)

(B) Describe with neat sketch surveyor's compass.

(05)

(C) Write short notes : (Attempt any two)

(05)

1. Differential levelling and Fly levelling
2. Back sight and fore sight
3. G.T.S. bench mark, permanent bench mark.

SECTION - II

Q-4 (A) Classify the building based on its structure. Also explain substructure of building.

(04)

(B) What do you mean by conservation of water? Explain methods of conservations of water and describe any one.

(03)

Q-5 (A) Draw a detail sketch of a cross section of 30 cm thick wall showing various components of building.

(06)

(B) Describe traffic control devices and explain informatory sign in detail.

(04)

(C) Write short note on buttress dam and weir.

(04)

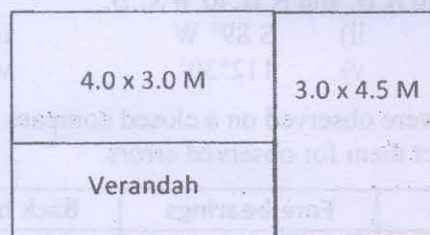
OR

Q-5 (A) Explain general principles of building planning. Describe roominess in detail.

(05)

(B) Draw a detail plan including various components of a building from given line diagram. (Use scale 1 : 50)

(05)



(C) What is the role of transportation in national development? Classify the different ways of transportation.

(04)

(P.T.O)

- Q-6 (A) Explain the functions of following (06)
- | | | |
|-----------------|------------|-------------|
| i. Foundation | ii. Plinth | iii. Lintel |
| iv. Weathershed | v. Roof | vi. D.P.C |
- (B) Write short note on B.O.T. Project. (03)
- (C) What do you mean by water losses? Describe the causes of water losses and explain any one in detail. (05)

OR

- Q-6 (A) Define infiltration, Evaporation, Aquifer (06)
- (B) Write short note on Canal regulation works (04)
- (C) Explain various design loads acting on a structure. (04)